Ionizing Radiation Detection and Nuclear Instrumentation [LPHYS2102]
Academics : Cocolios Thomas Elias, Cortina Gil Eduardo, Severijns Nathal



UCLouvain

Compton scattering

Laboratory report
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Bogaert L., Bouhalbane M. & Dartois R.

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Introduction

We present in this paper all the steps we followed in order to carry out the Compton experiment. Our report is divided into two parts. The first being the one where we characterize and calibrate our NaI detector by means of different radioactive sources. We talk about the resolution and the efficiency of our detector and we also have a look at the electronics that make it up. The second one is the part in which we do the actual Compton scattering and correlate our data with the theory. We explain how we obtain the data that allow us to get the differential cross section as a function of the angle which we will compare with the Klein-Nishina formula at the end.

Detector characterization for Compton experiment

This first part of the project involves calibrating the detector to be used. Once studied and fixed, this calibration will enable us to carry out the Compton experiment (see dedicated part). More concretely, this section on the characterization of the detector will firstly develop the experimental setup by talking about the sources and the detector used, as well as the detector electronics. Secondly, the calibration of the detector will be discussed, explaining how the measured energy peaks are adjusted, and also making the link between the detector's measurement channel and the energy to which it corresponds. Thirdly, the detector's performance will be developed in three points : energy resolution, efficiency as a function of distance and efficiency as a function of energy.

1 Experimental Setup

1.1 Sources and detector

On the one hand, for the entire detector calibration section, we use several known sources to obtain various references for their energy values. The characteristics of the sources are given in the table 1 below. The source used in the second part of the Compton experiment is different, and its details will be developed in a dedicated section.

Source	Half-life	Datum	Activity
Na-22	2.60 yr	05-2008	100 µCi
Co-60	5.27 yr	12-2010	1 µCi
Cs-137	30.07 yr	11-2010	5 µCi
Eu-152	13 yr	04-2010	11 µCi

TABLE 1 – Details of radioactive references sources used in the detector characterization.

On the other hand, all detections will be carried out using a single detector : a sodium iodide NaI(Tl) scintillation detector used for gamma counting. We chose this type of detector over others for several reasons :

— Fast response to radiation passage
— Large detection volume enabling higher efficiency for high E gamma ray
— operate at room temperature

Though scintillators have a worse resolution than HPGs detector, it is not an issue here because we are more interested in the efficiency than the resolution.

1.2 Electronics

The subsection dedicated to detector electronics presents the various modules involved in the signal processing chain, from the detection of a physical event by the detector to the production of an electrical signal that can be processed by the acquisition system.

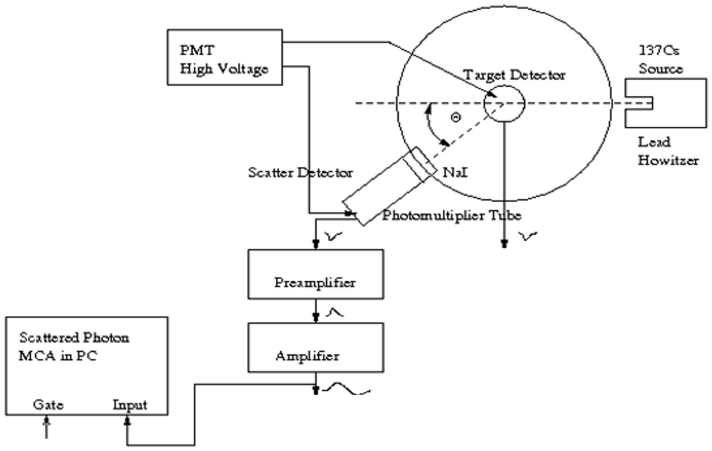


FIGURE 1 – The illustration shows the electronic chain used, starting from the source at the top right of the Compton scattering experiment (described in detail in Part 2). The schematic then follows the signal's path from the detector, through the various electronic processing modules, to its final acquisition by the acquisition system.

The figure (1) above shows the electronics that is used for the Compton experiment. We have a high voltage to power up the PMT in the NaI detector which will enable it to multiply the current produced by incident light. For the computer to be able to read the outgoing signal of the detector, we still need to amplify its amplitude by means of a pre-amplifier and an amplifier. Finally, there is the MCA to assign each pulse to a channel based on this energy by building a histogram : number of pulses with respect to pulse amplitude/energy. With this circuit, we can now analyze the energy distribution of detected radiation signals.

2 Detector calibration

This section first describes the numerical fit of the energy peaks, in order to derive usable information from the measured spectra. Next, the measured channel of energy peaks will be assigned to an energy. This is the preliminary stage in characterizing the detector, and in the subsequent section the performance of the detector will be investigated.

2.1 Fit of a peak

The detector records the spectrum of the source under study. Based on this, numerical processing is performed to derive information on energy peaks.
Here we take the example of Na22 to show the fit applied more generally to all peaks of each radioactive sources. Specifically, we make a linear fit model for the background and a Gaussian fit model for the signal :

$$y = Ae^{\frac{-(x-\mu)^2}{2\sigma^2}} \tag{1}$$

Where the parameters of the fit are A, μ and σ representing respectively the constant, the mean and the sigma. A linear background fit is also applied to account for the environment. The total fit of a peak is the Gaussian fit raised by the background fit.

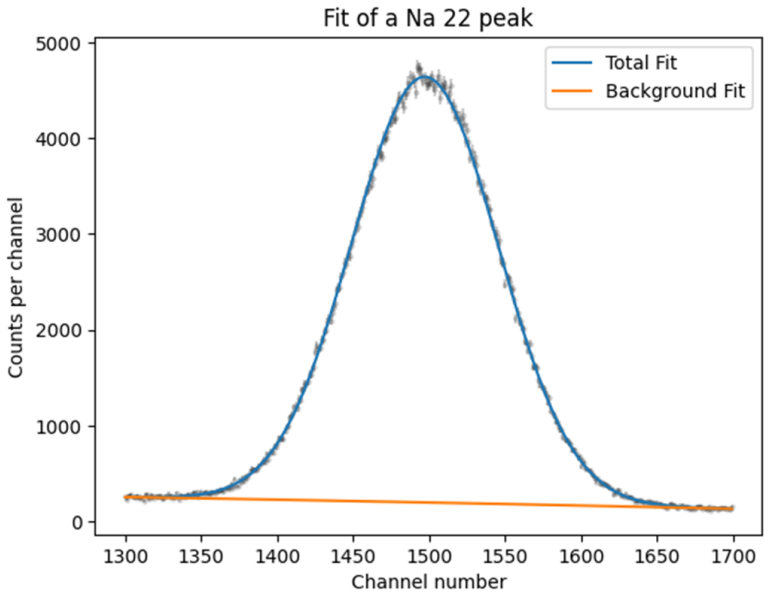


FIGURE 2 – Adjustment of a Na 22 peak using a Gaussian peak adjustment and a linear background. The χ^2 value of this total fit is $\chi^2 = 1.0$.

For the fit of an energy peak shown in the figure (2), the various fit parameters are listed in the table 2. This information is taken for each source, each peak energy and for several distances. This data will be useful to first link a peak detection channel (the mean μ) to the known peak energy. Then, they will be used to evaluate the detector's performance.

Symbol	**Name**	**Value**	**Error**
A	Constant	4.442e3	0.008e3
μ	Mean	1.49710e3	0.00008e3
σ	Sigma	48.32	0.07

TABLE 2 – Parameters of the total fit obtained for the 511 kev peak of Na 22 shown in figure (2).

2.2 Channel as a function of energy

In this section, we show what fit we used to assign a channel to each energy value. We know that specific sources emit characteristic energies. So these theoretical energies and these values are assigned to each channel where there is a peak in the spectrum.
The resulting plot is illustrated in figure (3) which has been modeled by the linear equation :

$$\text{Channel} = a \cdot \text{Energy} + b \tag{2}$$

The fit parameters obtained are $a = 2760.89 \pm 20.99$, $b = 66.26 \pm 17.60$ such that the dependency between the energy and the detection channel is then :

$$\text{Channel} = 2760.89 \cdot \text{Energy} + 66.26 \tag{3}$$

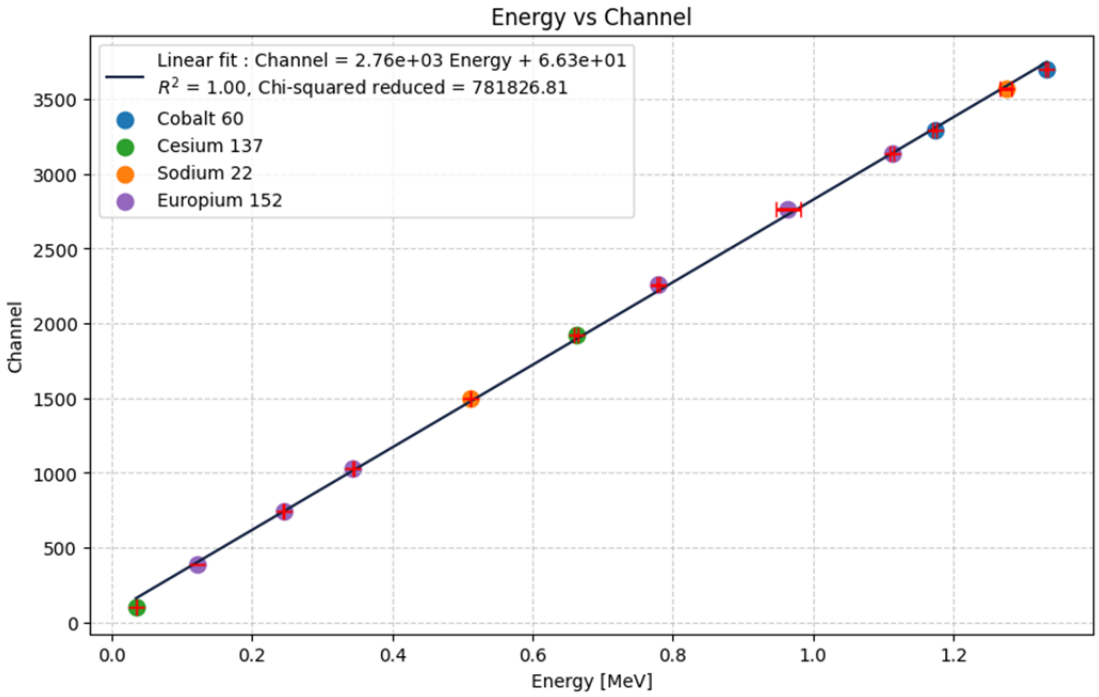


FIGURE 3 – Dependence between energy and channel measured by the detector

Thanks to this dependence between energy and channel, our detector is characterized and we can directly assign the recorded channel of any source to an energy by the equation (3).

The reduced chi-square is defined by the relation $\chi^2_{\text{red}} = \frac{1}{\nu}\sum_{i=1}^{n}\left(\frac{y_i^{\text{exp}} - y_i^{\text{th}}}{\sigma_i}\right)^2$, where ν is the number of degrees of freedom and σ_i are the uncertainties associated with each point. In general, for all the adjustments made in this project, when $\chi^2_{\text{red}} \approx 1$, this indicates that the model is consistent with the data given the errors. A $\chi^2_{\text{red}} < 1$ indicates an overestimation of uncertainties, while a $\chi^2_{\text{red}} > 1$ indicates larger than expected deviations, due either to a bad fit or to an underestimation of uncertainties. However, Figure (3) shows that the experimental points are perfectly aligned with the fitted line, with a linear regression coefficient of $R^2 = 1.00$. However, the values of χ^2_{red} obtained are abnormally high ($\chi^2_{\text{red}} = 781826{,}81$). This can be explained by small

differences between the data and the model, relative to very small uncertainties, which amplifies their contribution to χ^2. This mismatch suggests a significant underestimation of uncertainties, or a poor implementation of the chi-square formula in the adjustment code. This leads to a distorted interpretation of the chi-square, making it unrepresentative of the visual quality of the fit, as confirmed by the $R^2 = 1.00$.

3 Detector performance

Three analyses will be carried out to study the detector's performance : energy resolution, efficiency as a function of distance and efficiency as a function of energy.

3.1 Energy resolution

A very important feature of the detector is the energy resolution. It gives the precision with which the detector can measure the energy of the photons. It is given by this equation :

$$R = \frac{\text{FWHM}}{E} = 2.35\ \frac{\sigma}{E} \tag{4}$$

The FWHM, or *Full Width at Half Maximum*, corresponds to 2.35 σ for a Gaussian distribution. To verify this dependency, we plot the resolution as a function of energy and observe an inverse square-root relationship, which indicates that the energy degradation follows a statistical process. Thus, we use the following equation for the fit :

$$R = a + \frac{b}{\sqrt{E}} \tag{5}$$

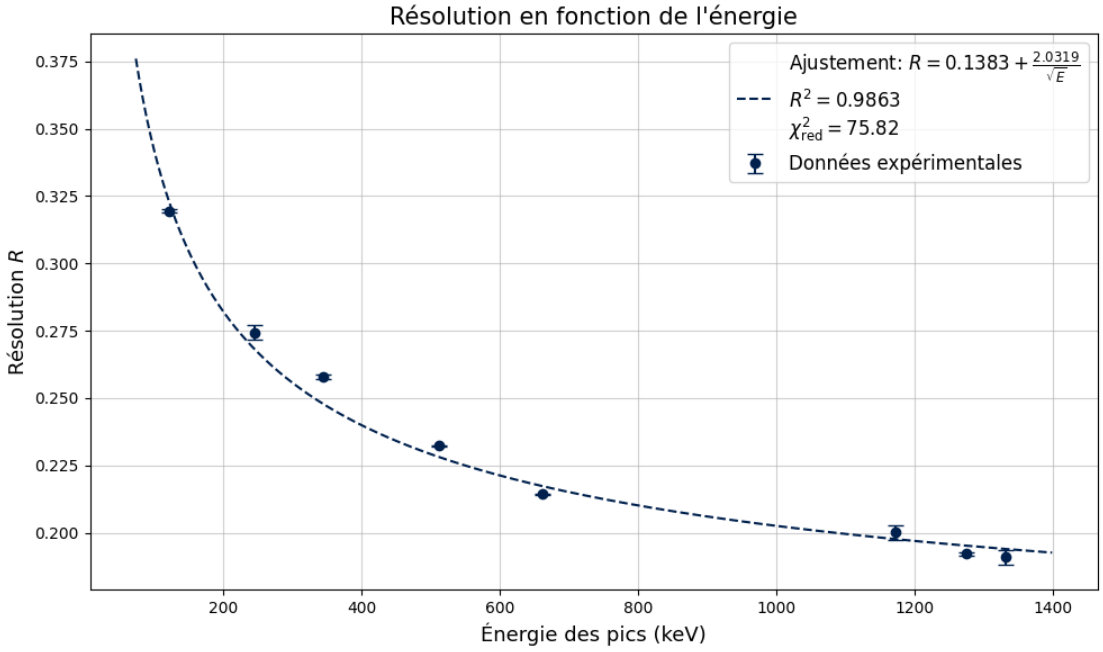


FIGURE 4 – Plot of the energy resolution as function of the energy

By adjusting the fit, equation (5), across the different experimental energy points, we obtain the fit parameter values $a = 0.1383 \pm 0.0006$ and $b = 2.0319 \pm 0.0113$. Where the parameter a is taking into account the statistical fluctuation and the parameter b has a value close to 2.35, as expected from the theoretical formula.
The figure (4) shows that the fitting curve follows the experimental data with a linear regression coefficient close to 1. However, the reduced chi-square value remains high, suggesting again an underestimation of uncertainties or an error in the fitting code.

3.2 Efficiency

An important factor to take into account in the characterization of our detector is its efficiency. Efficiency is the ratio between the number of events emitted and the number of events detected by the detector, and is therefore given by the formula :

$$\epsilon = \frac{N_{\text{detect}}}{N_{\text{emitted}}}$$

where

$$N_{\text{emitted}} = A \cdot I_\gamma \cdot \Delta t$$

With A the activity of the source given by $A = A_0 \cdot e^{-t/\tau}$ and I_γ the emission rate of a photon
We distinguish that efficiency depends on intrinsic efficiency and geometric efficiency, which is given by the solid angle :

$$\epsilon = \epsilon_{\text{intrinsic}} \cdot \epsilon_{\text{geometric}} = \epsilon_{\text{intrinsic}} \cdot \frac{1}{2}\left[1 - \frac{d}{\sqrt{d^2 + r^2}}\right] \tag{6}$$

where d is the distance between the source and the detector and r is the radius of the detector, which is 2.54cm.

3.2.1 Efficiency as a function of distance

It is possible to determine the efficiency based on distance. In this case, the distance d between the source and the detector becomes a variable in equation (6). This means that we need to take measurements for different sources at different distances : $d = 6, 7, 8, 10, 12, 14, 16, 18, 20, 25$cm.
To adjust the efficiency according to distance, we use equation (6). Since we do not know the intrinsic efficiency, we set it as a parameter a. Thus, by fitting the efficiency according to distance, we can experimentally find the value of the intrinsic efficiency by the value of the parameter a. The fit is then :

$$\epsilon = \frac{a}{2}(1 - \frac{d}{\sqrt{d^2 + r^2}}) \tag{7}$$

By adjusting this to experimental data that varies with distance, we obtain figure (5), where the parameter is $a = 0.0757 \pm 0.0055$.

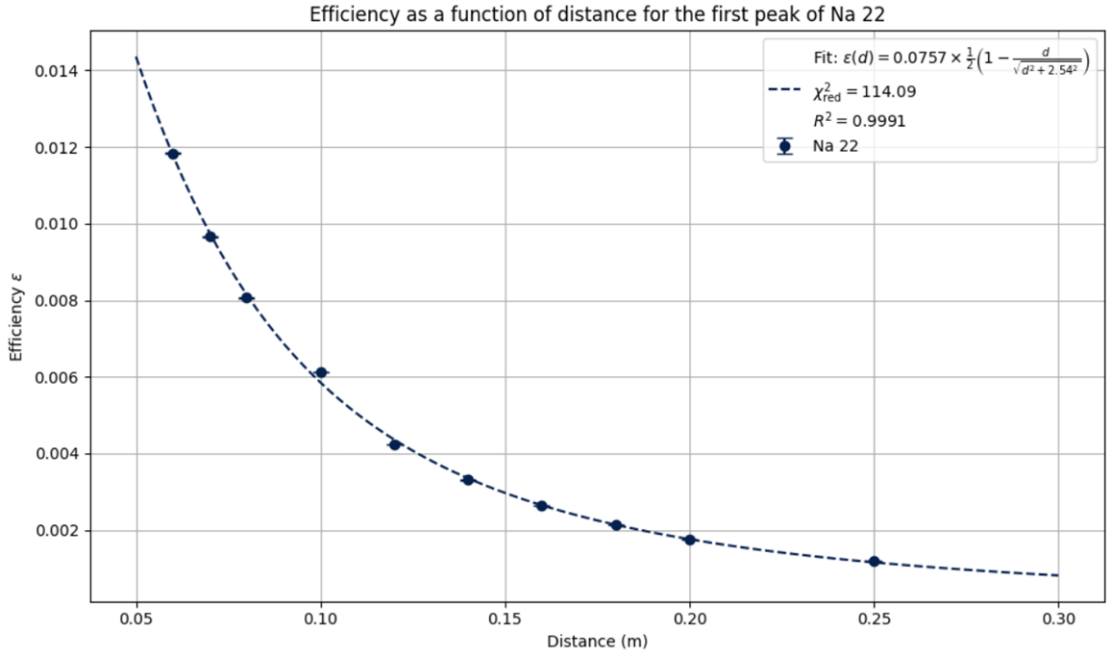


FIGURE 5 – Efficiency as function of the distance for the source ^{22}Na

The figure (5) only represents the experimental data for this fit taken at different distances for the sodium 22 source. However, the efficiency values as a function of distance for the different sources are shown below in table form.Tables 3, 4, and 5 show the efficiency as a function of distance for sodium-22, cobalt-40, and cesium-137, respectively.

Distance (cm)	ε_1 ($\times 10^{-3}$)	Error ε_1 ($\times 10^{-5}$)	ε_2 ($\times 10^{-4}$)	Error ε_2 ($\times 10^{-6}$)
6	20.250	2.7884	38.704	12.093
7	16.519	2.5140	31.975	10.988
8	13.793	2.2942	27.166	10.126
10	10.472	1.9957	21.168	8.9355
12	7.2337	1.6560	14.948	7.5064
14	5.6557	1.4631	11.556	6.5989
16	4.5063	1.3053	9.2206	5.8938
18	3.6396	1.1726	7.4666	5.3032
20	2.9910	1.0626	6.5959	4.9842
25	2.0476	0.8788	4.3919	4.0667

TABLE 3 – Efficiency as function of the distance for the source ^{22}Na

Distance (cm)	ε_1 ($\times 10^{-3}$)	Error ε_1 ($\times 10^{-5}$)	ε_2 ($\times 10^{-4}$)	Error ε_2 ($\times 10^{-6}$)
6	3.4474	3.1174	30.492	29.312
7	2.6268	2.7200	24.602	26.000
8	2.0972	2.4298	19.152	23.217
10	1.5112	2.0620	14.278	20.042
12	1.1340	1.7859	9.8343	16.629
14	0.82920	1.5269	7.6060	14.623
16	0.66693	1.3692	5.7634	12.728
18	0.62113	1.3214	4.9127	11.751
20	0.51137	1.1989	3.6079	10.069
25	0.26438	0.86192	2.3757	8.1703

TABLE 4 – Efficiency as function of the distance for the source ^{60}Co

Distance (cm)	ε ($\times 10^{-3}$)	Error ε ($\times 10^{-5}$)
6	7.0460	1.0187
7	5.7978	0.92354
8	4.7521	0.83568
10	3.3596	0.70216
12	2.4732	0.60219
14	1.9143	0.52965
16	1.5282	0.47313
18	1.2409	0.42628
20	1.0223	0.38688
25	0.69849	0.31974

TABLE 5 – Efficiency as function of the distance for the source ^{137}Cs (only the second peak)

3.2.2 Efficiency as a function of energy

Another study of efficiency can be carried out : efficiency as a function of energy. To do this, we need to know the intrinsic efficiency using the parameter a obtained from the fit (6) in figure (5). By fixing the distance d, taking the different energy values of the sources and fitting the experimental data using the following equation :

$$\epsilon = a + b\ log(E) + c\ log^2(E) \tag{8}$$

The following adjustment is obtained from the experimental data :

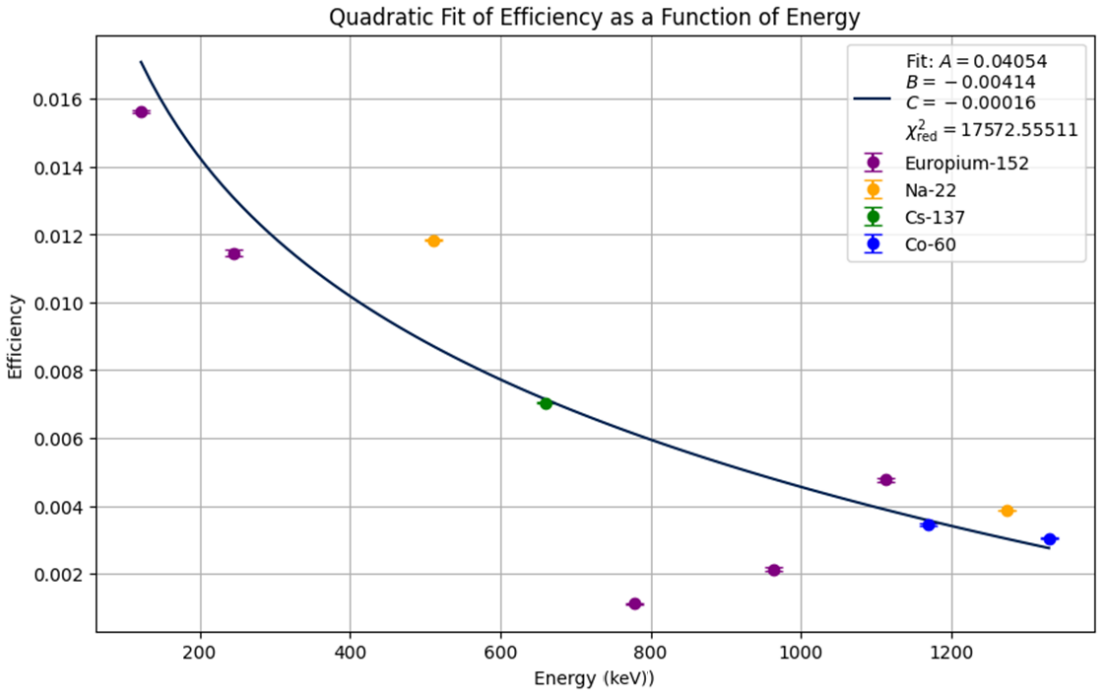


FIGURE 6 – Efficiency as function of energy

with the parameters : $a = 0.04054 \pm 0.00056$, $b = -0.00414 \pm 0.00018$ and $c = -0.00016 \pm 0.00001$. We can see from the graph, the points follow the fit given by the equation, except for a few points such as Na 22 or certain europium points which should have been close to the curve. We also notice that the value of χ^2 is high for the same reason as explained before.

4 Summary of the first part

To sum up, in this first part we did several things. We associated a channel with an energy thanks to the gaussian fit of the Na 22, Co 60, Cs 137 and Eu 152 peaks. We calibrated our NaI detector by looking at :

1. Energy resolution
2. Efficiency as a function of distance
3. Efficiency as a function of energy

This characterization of the detector with performance analysis will be useful in the second part of this report on the Compton effect since the same detector will be used.

Compton experiment

This second part of the project is about carrying out the actual Compton experiment with the aim of measuring the differential cross section for different angles θ. To do that, we now use exclusively Cs 137 as a source of photon. We still have our NaI detector we have characterized but we also put a aluminium target between the source and the detector and a rail to rotate the detector as you can see in figure (9).
This section on the Compton experiment is divided into three distinct parts. First, we will explain what Compton scattering is and how the position of the energy peak changes. Next, we will explain the experimental set-up, going on to explain the usefulness of the target, the shift of the peak at different angles, the theoretical and measured energy values and also verify the mass of the electron. Finally, we'll develop the differential cross section at various points : theoretically (Klein-Nishina equation) and experimentally with the results we obtained.

5 Introduction to Compton scattering

In the Compton scattering, we have a photon emitting source made of Cesium 137 at an energy of $E = \frac{h\nu}{c}$ that is going to hit a target thus interacting with the electrons of the matter. The principle is illustrated in figure (7). From this interaction and by conservation of momentum and energy, the initial photon and electron get scattered and both lose energy in the process. The energy of the photon after being scattered is such that $E' = \frac{h\nu'}{c} < E$.

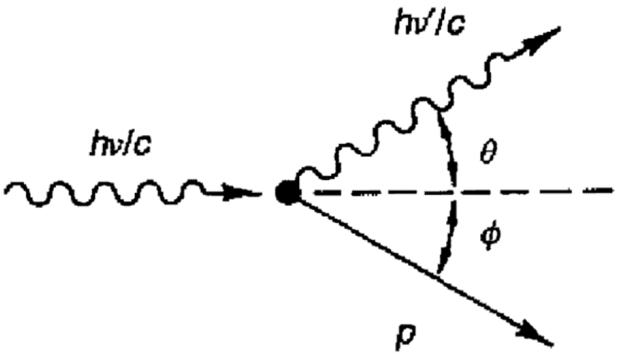


FIGURE 7 – Principle of the compton effect, which results from the scattering of an incident photon by a free electron. By conservation of momentum and energy, the photon transfers a quantity of its energy to the electron, which then acquires a non-zero velocity. The scattered photon then has a lower energy than the initial photon.

More precisely, by theoretical calculus of momentum and energy conservation, the outgoing final-state photon energy $E_{\gamma'}$ is given by :

$$E_{\gamma'} = \frac{E_\gamma}{1 + \left(\frac{E_\gamma}{m_e c^2}\right)(1 - \cos\theta)} \tag{9}$$

where E_γ is the initial photon energy and θ is the diffusion angle. In our case, we take $E_\gamma^{\mathrm{Cs}^{137}} = 661.657\,\mathrm{keV}$ and $m_e c^2 = 511\,\mathrm{keV}$.

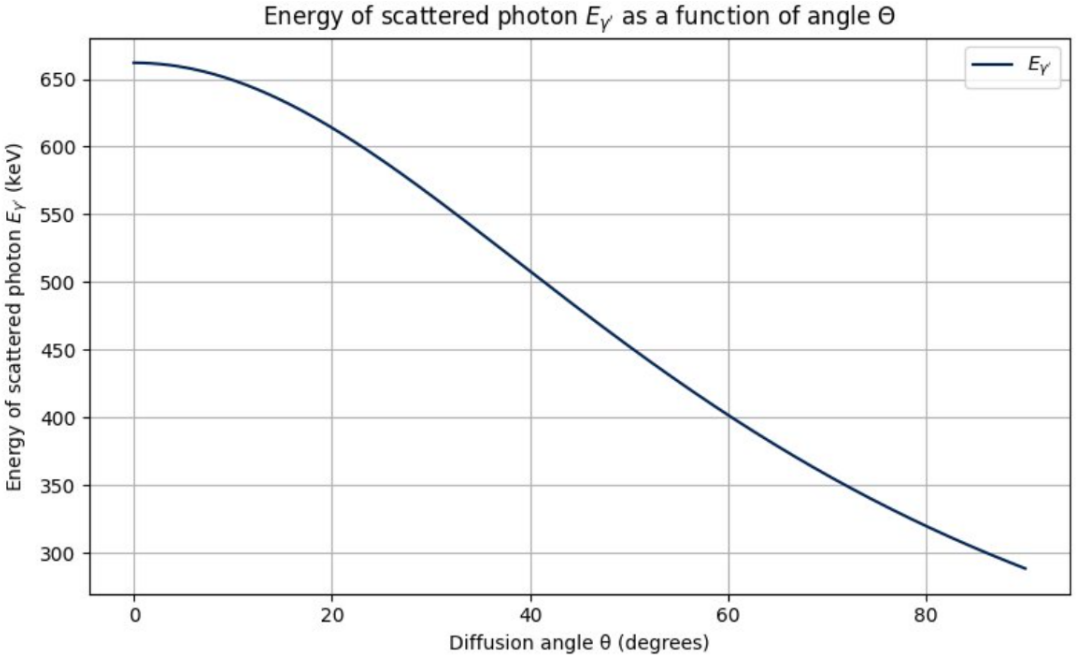


FIGURE 8 – Plot the theoretical energy curve of the scattered photon as a function of angle θ. We see that the energy decreases as angle increases

Equation (9) tells us that the energy of the outgoing photon must decrease when the angle θ increases as we can see in figure (8).

6 Experimental material

This section begins by showing the laboratory's experimental setup and giving information on the source used for the measurements that follow. Furthermore, the usefulness of the target is discussed, as well as the experimental shift in peak energy as a function of angle. Next, we compare theoretical and measured energy values for different scattering angles. Finally, a small verification is made to determine the electron mass from the energy values.

6.1 Experimental setup

In a effort to detect the scattered photons, we move our scintillation detector around the target with an angle θ between 0° and 90°. From the computer communicating with our detector, we get the number of counts of scattered photons for a specific angle with respect to the channels which can be linked to a certain energy by the equation (3).

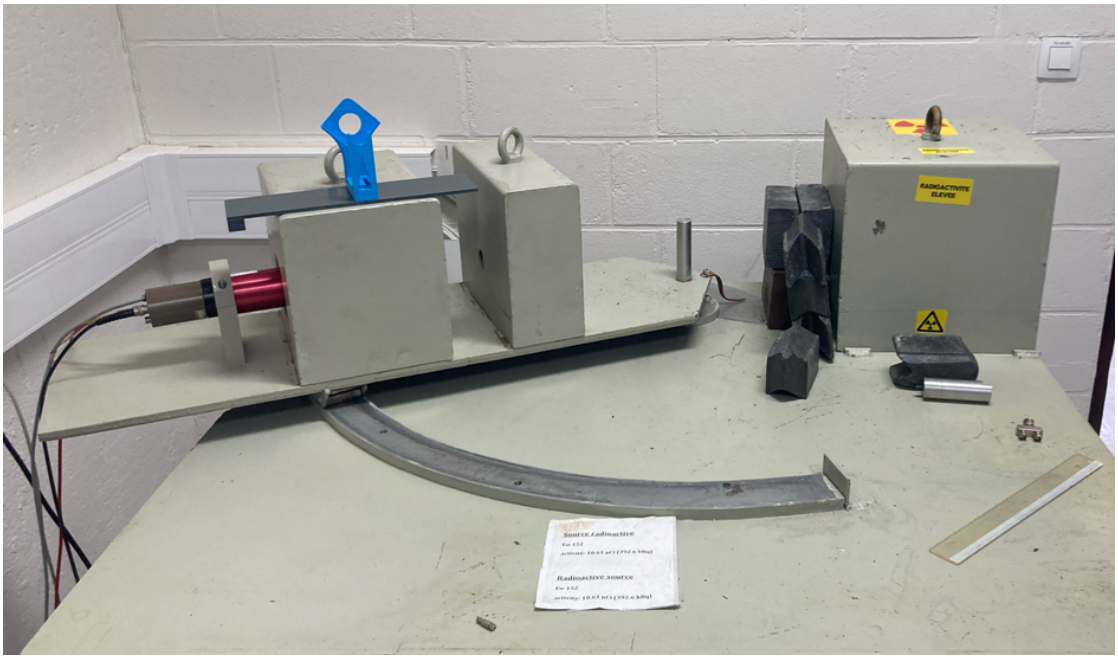


FIGURE 9 – Setup of the experiment in the lab. From left to right : (1) the NaI detector characterized in the first part of this report, surrounded by lead to reduce noise. This part of the experiment is mounted on a rail so that the angle of detection can be varied around the fixed target. (2) The aluminum target in the center scatters the photons coming from the source. (3) The source (here shielded by lead when no detection is performed) of Cs 127. The source data are given in the table (6).

The source used throughout the Compton experiment is described in the following table (6) :

Source	**Half-life**	**Datum**	**Activity**
Cs-137	30.07 yr	18-04-1978	6 mCi

TABLE 6

6.2 Usefulness of the target

The question may arise as to the usefulness of the target in Compton's experiment. The figure (10) illustrates the counts as function of the channel with and without the target for $\theta = 50°$ and for a time of 840 minutes. We notice that the peak around 1400 is not there for the plot without the target. Therefore we note the necessity to set the target in order to highlight the Compton effect because without the aluminum target, there's simply no Compton effect.

6.3 Shift of the peak at different angles

We can also experimentally verify the equation (9) illustrated in the figure (8), which tells us that the energy of the scattered photon detected is smaller as the scattering angle increases. Experimentally, it can be seen in figure (11) that as the angle θ increases from 50° to 70° during the same time of 840 minutes. The peak shifts to the left meaning that the energy $E_{\gamma'}$ decreases as well. Hence equation (9) for the position of the peak energy is verified by experiment.

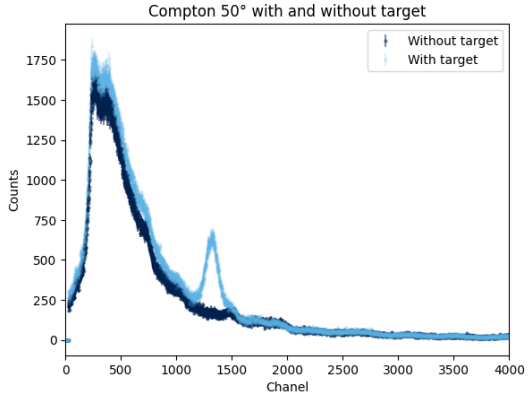


FIGURE 10 – Visualization of the usefulness of the target for the same angle and time. Without target, there is no scattered photon.

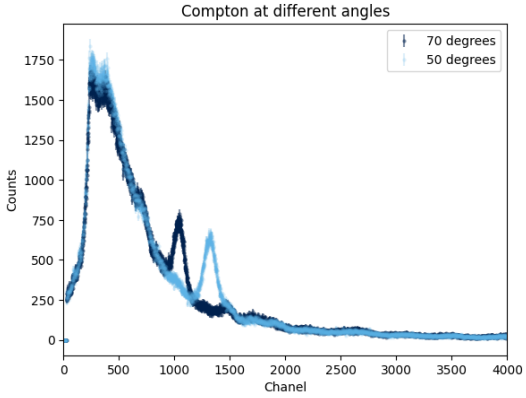


FIGURE 11 – Visualization of the position of the energy peak predicted by conservation of momentum and energy. The energy of the scattered photon is lower as the scattering angle increases.

6.4 Theoretical and measured energy values

Theoretically, energies are measured by the equation (9) for the different angles. Experimentally, we apply the same type of fit as developed in the 2.1 section to measure the peak detection channel. Then, using the equation (3), we find the value of this theoretical energy.
The table (7) shows the energy values calculated theoretically and measured experimentally, as well as the error on the measured energy. The data were calculated for angles from 20° to 90° in 10° steps. The first two angles, 0° and 10°, are not included in the data, because at these small angles, our detection is completely saturated by photons coming directly from the source and not being scattered. Indeed, it's these scattered photons that interest us for the Compton scattering experiment.

Angle (°)	**Peak Channel**	$\mathbf{E'_{theory}}$ (keV)	$\mathbf{E'_{measured}}$ (keV)	**Error** $\mathbf{E'_{measured}}$ (keV)
20	1760.0	613.73	613.48	7.90
30	1609.0	563.84	558.78	7.66
40	1475.8	507.82	510.54	7.46
50	1320.1	452.40	454.14	7.25
60	1175.1	401.63	401.62	7.07
70	1041.6	357.27	353.27	6.92
80	950.9	319.64	320.42	6.82
90	850.6	288.33	284.09	6.73

TABLE 7 – Table showing theoretical and measured energy values for different diffraction angles.

In order to compare these values, the measured energy can be plotted against the theoretical energy. If experimental measurements are in good agreement with theoretical values, the points should line up on the ideal straight line of equation y = x. Any deviation from this line reflects a difference between experimental results and theoretical predictions. This plot is illustrated in figure (12).

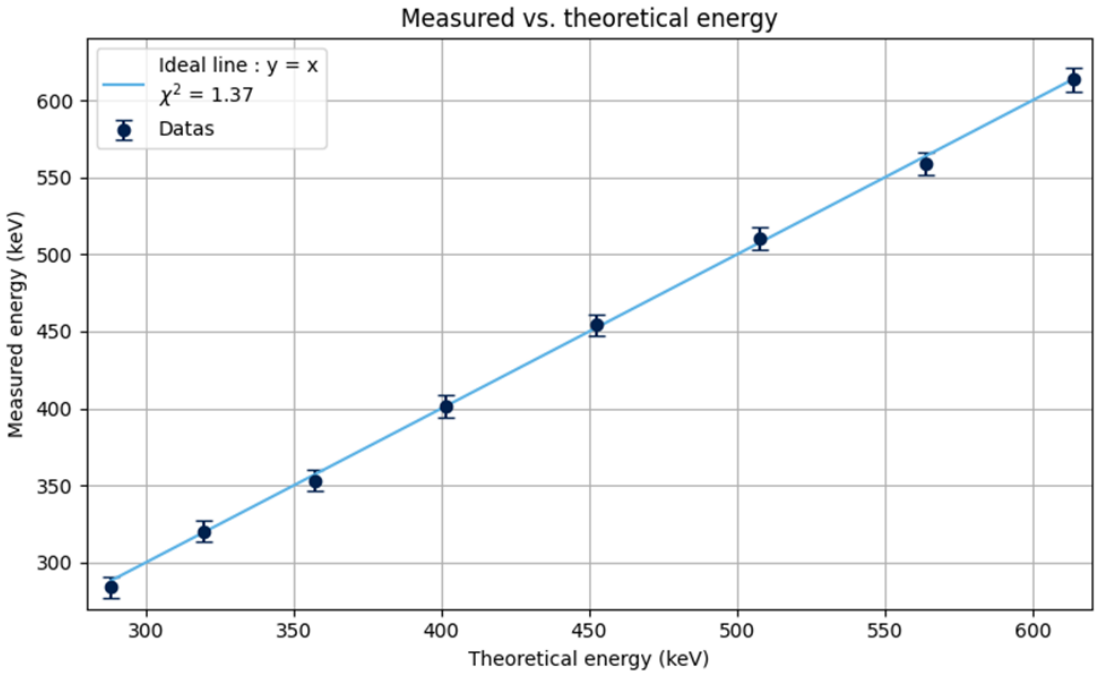


FIGURE 12 – Graphical representation of measured energy values in relation to theoretical. If experimental measurements are in good agreement with theoretical values, the points should line up on the ideal straight line of equation y = x.

6.5 Verification : mass of the electron

A simple verification consists of using the values of the theoretical energy and the experimentally measured energy to find the mass of the electron (511 keV) from the equation (9). To do this, we can rewrite equation (9) in the following form :

$$\frac{1}{E_{\gamma'}} = \frac{1}{m_e c^2}(1 - \cos\theta) + \frac{1}{E_\gamma} \tag{10}$$

In this equation, the slope of the line is equal to $\frac{1}{m_e c^2}$. Then by doing a linear fit of the experimental data according to the relationship $\frac{1}{E_{\gamma'}}$ as a function of $(1-\cos\theta)$, we obtain the following equation :

$$\frac{1}{E_{\gamma'}} = 2.0199(1 - \cos\theta) + 1.48917 \tag{11}$$

The coefficients of the line fit are $a = 2.0199 \pm 0.0431$ keV^{-1} et $b = 1.4891 \pm 0.0104$ keV^{-1}. By comparing this equation to (10) we identify the slope as $\frac{1}{m_e c^2} = 2.0199$. This allows us to determine the electron mass $m_e c^2 = 0.495 \pm 0.011$ Mev. This value is quite close to the expected value of 511 keV, with a relative error of 3%.

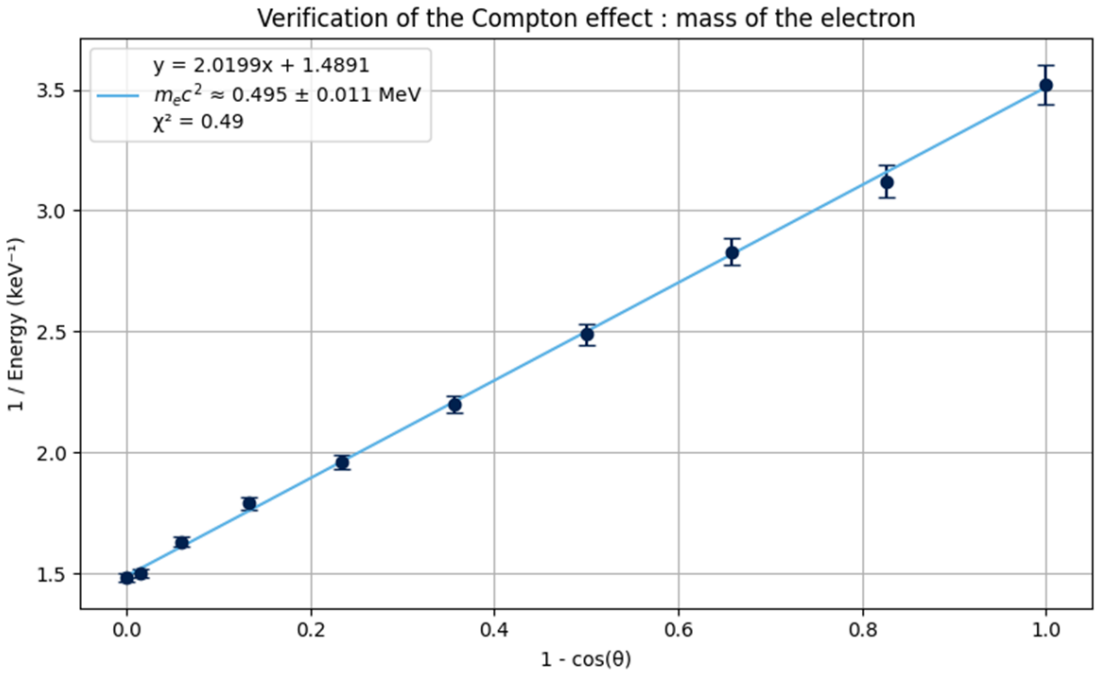


FIGURE 13 – Verification of the equation (9) obtained by conservation of momentum and energy. Knowing the energy values, it is possible to find the mass of an electron. The expected mass is 511 keV, while the value obtained experimentally is $m_e c^2 = 495 \pm 11$ kev.

7 Differential cross section

This section concerns the calculation of the differential effective cross section. To do this, we will develop the subject in three sub-sections : the first one deals with the theory behind the Klein-Nishina formula, which enables us to make the theoretical adjustment to the experimental data. The second part explains how to obtain experimental values for the differential cross section of Compton scattering. Finally, the third part discuss the results obtained and the theoretical fit of the Klein-Nishina differential cross section on the experimental scattering data.

7.1 Klein-Nishina :

The Klein-Nishina formula is used to express the differential cross section of photons scattered from a single free electron. It accounts for relativistic effects as opposed to the Thomson formula which is only valid for low energy photon.

7.1.1 Theoretically

Theoretically, Klein Nishina's formula is expressed as :

$$\frac{d\sigma}{d\Omega} = r_0^2 \cdot \frac{1+\cos^2\theta}{2} \cdot \frac{1}{[1+\gamma(1-\cos\theta)]^2} \cdot \left[1 + \frac{\gamma^2(1-\cos\theta)^2}{(1+\cos^2\theta)\,[1+\gamma(1-\cos\theta)]}\right] \tag{12}$$

In this expression (12), on the one hand, r_0 represents the classical electron radius such that $r_0 = \frac{e^2}{4\pi\epsilon_0 mc^2} = 2.82 \times 10^{-13}$cm. On the other hand, γ, is the ratio of the incident photon's energy to the electron's rest energy and is $\gamma = \frac{h\nu}{mc^2}$. This equation is plotted in figure (14).

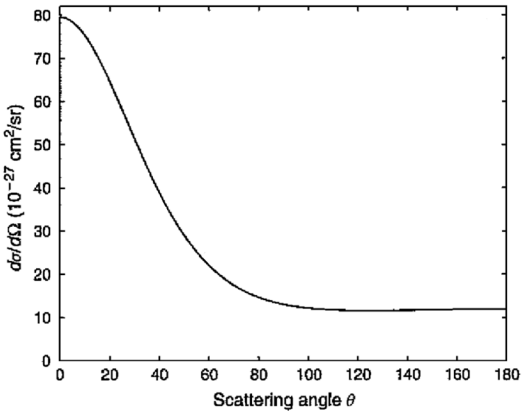


FIGURE 14 – Illustration of the Klein-Nishina cross section. The differential cross section is plotted as a function of the scattering angle.

7.1.2 Experimentally

The next step is to measure the differential cross section for different scattering angle. To obtain this value we use the following formula for the differential cross section :

$$\frac{d\sigma}{d\Omega} = \frac{\text{yield}}{(d\Omega)NI_0t} \tag{13}$$

We need to calculate each term in this equation (13). First, the detector solid angle is given by :

$$d\Omega = \frac{4\pi}{2}\left[1 - \frac{d}{\sqrt{d^2+R^2}}\right] = 6.28 \times 10^{-3} \text{ sr},$$

where d is the distance from the target to the source and R is the radius of the detector.
Next, the flux density of the photon beam at the target is given by I_0 :

$$I_0 = \frac{AI_\gamma}{4\pi r^2} = 1703.7 \text{ photons/cm}^2\text{-s},$$

I_0 uses the following information : r is the distance between the source and the detector. $I_\gamma = 0.851$ is the emission rate. For the activity, the Cs 137 source has an activity of $220MBq$ in 1978 therefore $A = 220 \text{ MBq} \cdot e^{-\frac{\ln(2)}{30.17}\cdot 47} = 74.8 \times 10^6 Bq$.
Finally, the total number of electrons in the target is express as :

$$N = \pi\left(\frac{d}{2}\right)^2 h\rho\frac{N_0}{A}Z$$

where

$$\begin{aligned}
\rho &= \text{density of aluminum} = 2.7\ \text{g/cm}^3 \\
N_0 &= \text{Avogadro's number} = 6 \times 10^{23} \\
A &= \text{atomic weight of aluminum} = 27 \\
Z &= \text{atomic number of aluminum} = 13 \\
d &= \text{diameter of target} = 2.5\ \text{cm} \\
h &= \text{height of target} = 10.2\ \text{cm}
\end{aligned}$$

So, by putting the value of $d\Omega$, I_0, N into the equation (13). We can construct the table (8) :

Angle (°)	**Canal du pic**	**Temps (min)**	$\frac{d\sigma}{d\Omega}$ (10^{-27} cm^2/sr)	**Erreur sur** $\frac{d\sigma}{d\Omega}$ (10^{-27} cm^2)
20	1760.0	120	21.18	2.20
30	1609.0	180	17.52	1.82
40	1475.8	240	14.98	1.56
50	1320.1	840	11.39	1.18
60	1175.1	840	8.41	0.89
70	1041.6	840	8.34	0.87
80	950.9	2880	7.51	0.78
90	850.6	2880	6.03	0.63

TABLE 8 – Experimental values of Compton scattering

7.1.3 Results and discussion

This final sub-section discusses the Klein-Nishina differential effective cross section shown in section 7.1.1 through the experimental data developed in section 7.1.2.
By fitting the experimental data in the table (8) by the Klein-Nishina equation with an amplitude factor **A** to adapt the scale of the model to the data (equation (14)). This fit through the data gives us the figure (15) below.

$$\frac{d\sigma}{d\Omega} = \mathbf{A} \cdot r_0^2 \cdot \frac{1+\cos^2\theta}{2} \cdot \frac{1}{\left[1+\gamma(1-\cos\theta)\right]^2} \cdot \left[1 + \frac{\gamma^2(1-\cos\theta)^2}{(1+\cos^2\theta)\left[1+\gamma(1-\cos\theta)\right]}\right] \tag{14}$$

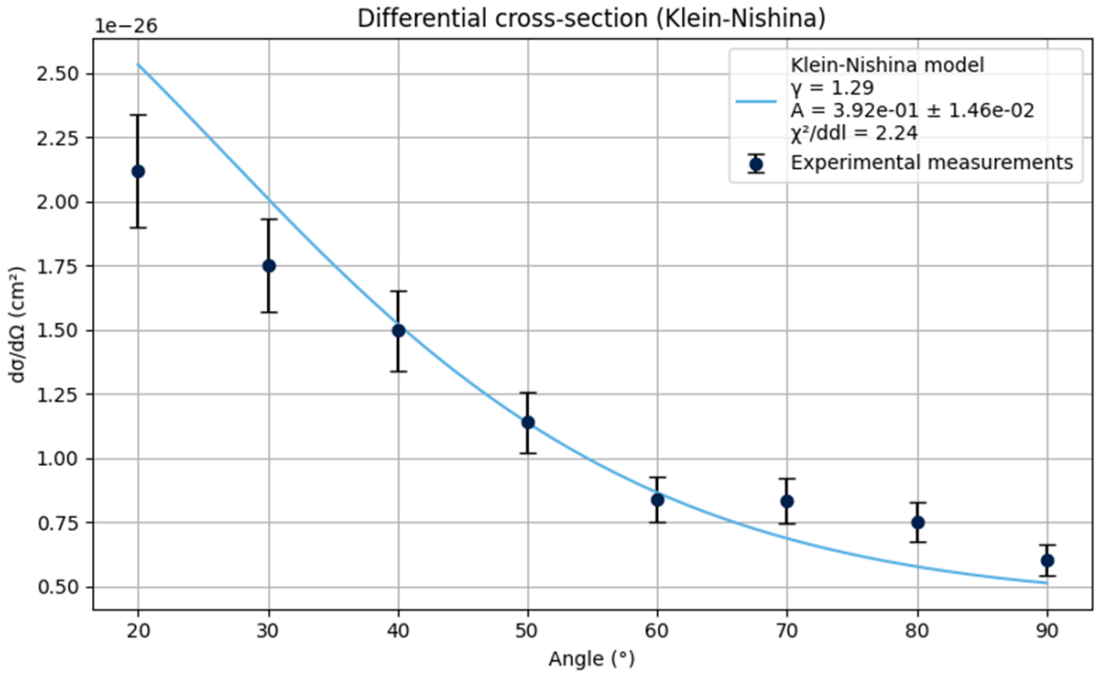


FIGURE 15 – Adjustment of the theoretical Klein-Nishina curve (equation (14)) with an amplitude factor value $A = 0.3924 \pm 0.0146$ through measured table points (8) obtained from equation (7.1.2).

Throughout the analysis of the differential cross section, we have neglected the influence of the detection efficiency in equation (13), even though the efficiency should in principle be taken into account. This is due to the low precision on the efficiency values derived from the fit equation (8) which introduce significant uncertainty into the position of the experimental points. Integrating these values degrades rather than improves the reliability of the fit. Despite this approximation, the figure (15) shows that the curve obtained by fitting roughly follows the plot of the measured points. However, there are some notable divergences : the measured values are too low at small angles and too high at large angles. This suggests that a multiplicative factor, probably due to efficiency, has not been taken into account.

Conclusion

After characterizing our detector and evaluating its performance in the first part, in particular its energy resolution, efficiency as a function of distance and efficiency as a function of energy, we proceeded to the second part : the Compton experiment.
This second part, building on the first, presented the Compton effect including the formula for the energy of scattered photons derived from energy conservation, as well as a description of the experimental setup. This allowed us to understand the role of the scattering target, observe the shift of the energy peak towards lower energies as the scattering angle increases, and determine the mass of the electron experimentally. The last part focused on studying the differential cross section by fitting the Klein-Nishina formula to the measured scattering data. A clear improvement in the experimental results lies in the poor fit of the efficiency as a function of energy done in the first part, which affects the accuracy of the fit of the differential cross section data to the Klein-Nishina curve.

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